



Facets of Disparate Impact: Evaluating Legally Consistent Bias in Machine Learning

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Abstract

Leveraging current legal standards, we define bias through the lens of marginal benefits and objective testing with the novel metric “Objective Fairness Index”. This index combines the contextual nuances of objective testing with metric stability, providing a legally consistent and reliable measure. Utilizing the Objective Fairness Index, we provide fresh insights into sensitive machine learning applications, such as COMPAS (recidivism prediction), highlighting the metric’s practical and theoretical significance. The Objective Fairness Index allows one to differentiate between discriminatory tests and systemic disparities.

CCS Concepts

• **Social and professional topics** → **Computing / technology policy**; **User characteristics**; *Race and ethnicity*; *Gender*; *Age*; • **General and reference** → **Metrics**.

Keywords

fairness, bias, bias metric, protected classes, legal policy, discrimination law, objective fairness index, disparate impact

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1 Introduction

In the 2010s, machine learning took major leaps forward in everyday life. However, we soon realized that bias is rampant in machine learning models. One of the most prominent examples regarding COMPAS, which deals with Americans’ civil rights in the United States justice system [3, 12]. Researchers are actively working to mitigate bias in general in a myriad of ways [5, 6, 8–11, 13, 15–19, 21–23]. Despite being the cornerstone of many proposals, definitions of bias remain inconsistent [6, 20]. Due to various philosophies and situations, there is no consensus on a formal definition. We focus on the legal context, namely the disparate impact metric (DI). DI is useful as a flag or reparative measure, however it is insufficient in objective-testing laws.



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To address this gap, we introduce the Objective Fairness Index (OFI), extending our work in [4]. We formalize OFI by defining bias as the difference between marginal benefits, derived via the lens of objective testing. In law, objective testing is the philosophy that correct assessments found via rational, related tests are not discriminatory. Additionally, we provide applications to two critical bias scenarios: COMPAS and Folktale’s Adult Employment dataset.

The principle of objective testing in non-discrimination laws is recognized not only in the US but worldwide. According to [7], 34 out of 35 surveyed countries (including Spain, Germany, Switzerland, Turkey, and Romania) have such laws in their national legislation. Going beyond the precedents found in the U.S. and the 34 other countries studied in [7], we find that South Africa (Employment Equity Act), Canada (Public Service Employee Relations Commission v British Columbia Government Service Employees’ Union), and Australia (Fair Work Act) uphold similar legal standards. Our work with the Objective Fairness Index aims to establish an agreeable and naturally interpretable definition of bias, grounded in these comprehensive legal frameworks and precedents.

We make four key contributions in this work:

- We substantially support the Objective Fairness Index with insights from the legal literature.
- We show that the disparate impact metric lacks the context of objective testing, which is integral in OFI.
- We bring novel insights into two popular ML fairness applications: COMPAS (predicts recidivism) and Folktale’s Adult Employment (predicts employment).
- We open source our code here¹. The code gives an efficient solution to handling many confusion matrices, creates the empirical results, and analyzes OFI’s behavior.

2 Limitations of Disparate Impact

This section gives context to the founding of the Objective Fairness Index by reviewing the Supreme Court of the United States’ decision that spurred disparate impact and a following precedent. Under objective-testing laws, we show that OFI is better than DI in identifying discriminatory tests.

2.1 Legal Foundations of Disparate Impact

Disparate impact’s significance comes from United States labor laws regarding the biased treatment of protected groups. In *Griggs v. Duke Power Co.*, 401 U.S. 424 (1971), the Supreme Court unanimously agreed that employers must prove that any test or assessment used must be related to job performance (objective). Written by Chief Justice Burger, the decision says that a lower admittance

¹<https://github.com/jarrenbr/objective-fairness-index>

rate of a protected class warrants further investigation—even if such discrimination may be unintentional. Since this precedent was passed from a lawsuit under Title VII of the Civil Rights Act of 1964, the Court’s ruling extends to decision-making whenever protected classes are concerned.

To summarize this ruling, employers must have objective and fair tests (neutral in form) and should not inadvertently advantage a protected class (neutral in impact). Should there be a disparate impact, the employer is subject to scrutiny and bears the burden of proof that the test is objective and “consistent with business necessity”. The disparate impact bias metric is defined in Definition 2.1. Where $P = FN + TP$ denotes the amount of positive labels, $\hat{P} = FP + TP$ denotes the amount of positive predictions, $N = FP + TN$ denotes the amount of negative labels, and $\hat{N} = FN + TN$ denotes the amount of negative predictions.

DEFINITION 2.1. *Disparate impact compares the rates of positive outcomes between two groups using ratios.*

$$DI \triangleq \hat{P}_i / n_i \div \hat{P}_j / n_j \quad (1)$$

The implication of Equation (1) is that $DI = 1$ suggests no bias. However, the real world can have small and non-representative sampling, causing unbiased decisions to be $DI \neq 1$. As such, a Four-Fifths Rule (80% Rule) has been established in law as an initial screening tool to identify potential discrimination (see 29 CFR § 1607.4 - Information on impact) [2]. This rule suggests that if $DI > 5/4$ then group i has positive bias, if $DI < 4/5$ then there is positive bias for j , otherwise DI finds no bias.

In *Griggs v. Duke Power Co.*, the Court goes on to note that Congress does not intend to require businesses to give jobs to unqualified candidates, and that context should still be considered, emphasizing that truly objective assessments are not discriminatory. However, because disparate impact lacks situational context, it cannot capture this legal nuance.

Disparate impact has been influential enough for employers to discard assessments that fail the heuristic to prevent any federal investigation or social uprising, regardless of the assessments’ objectivity. *Ricci v. DeStefano*, 557 U.S. 557 (2009) is the first of these cases to be argued in the Supreme Court when New Haven, Connecticut dismissed objective assessments that benefited the white and Hispanic firefighters over other protected classes. The Court ruled in the firefighters’ favor—that omitting an objective performance test solely due to differing acceptance rates (disparate impact) is a violation of the rights of those who correctly benefit from such a test. This provides us with a use case where equal positive prediction rates (which became zero after the test’s omission) are discriminatory.

2.2 Defining Benefit and Bias

We define bias as a function of what happened to the group (benefit) and what should have happened to the group (expected benefit). An objective test will show that benefit equals expected benefit. Consistent with common understanding and legal precedents, we define benefit as an advantage that is gained from a decision or action, such as hiring from a machine-learning prediction. We achieve formal definitions of these terms using binary confusion matrices, allowing generalization to situations where information such as

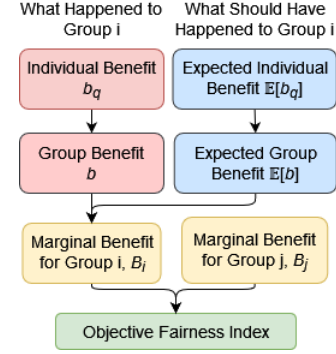


Figure 1: Objective Fairness Index’s conceptual framework.

specific features cannot be used due to privacy, legal protection, or information loss.

We define an individual benefit in Definition 2.2 to help describe a group’s benefit in Definition 2.3. We assume the positive prediction is the beneficial prediction in all definitions.

DEFINITION 2.2. *An individual q ’s benefit b_q is the individual’s prediction. $b_q = 1$ if predicted positive, otherwise 0.*

DEFINITION 2.3. *A group’s benefit is the arithmetic mean of all individual benefits within the group.*

$$b = \frac{1}{n} \sum_{\forall b_q} b_q = \frac{1 \cdot \hat{P} + 0 \cdot \hat{N}}{n} = \frac{TP + FP}{n} \quad (2)$$

The Objective Fairness Index also integrates the expected benefit of an individual (Definition 2.4) and the group’s expected benefit (Definition 2.5) by using the actual labels.

DEFINITION 2.4. *An individual’s expected benefit $\mathbb{E}[b_q]$ is their actual label. $\mathbb{E}[b_q] = 1$ if labeled positive, otherwise 0.*

DEFINITION 2.5. *A group’s expected benefit is the arithmetic mean of all individual expected benefits within the group.*

$$\mathbb{E}[b] = \frac{1}{n} \sum_{\forall b_q} \mathbb{E}[b_q] = \frac{1 \cdot P + 0 \cdot N}{n} = \frac{TP + FN}{n} \quad (3)$$

Formally defining benefit and expected benefit allows a counterfactual test of “what happened” versus “what should have happened”. As mentioned, this is of paramount importance in law. In *Ricci v. DeStefano*, the city of New Haven omitted their test solely based on the ratio of benefits to be obtained by each protected class of firefighters. However, this discriminates against the firefighters qualified by this objective test. Pursuant to the Supreme Court’s ruling, we present a solution by scoring this group’s treatment as the “marginal benefit” ($\mathcal{B}$ in Definition 2.6).

DEFINITION 2.6. *Marginal benefit ($\mathcal{B}$) is the difference between a group’s benefit and expected benefit.*

$$\mathcal{B} = b - \mathbb{E}[b] = (FP - FN) / n \quad (4)$$

This marginal benefit score suggests: $\mathcal{B} < 0$ an unjust lack of benefit; $\mathcal{B} = 0$ an appropriate amount of benefit; $\mathcal{B} > 0$ an unjust surplus of benefit. With marginal benefit defined, we now formalize

Table 1: Comparison of OFI, DI, and Law (Ricci v. DeStefano) Across Different Scenarios

Group	Scenario	TP	FN	FP	TN	n	b	$\mathbb{E}[b]$	$\mathcal{B}$	OFI: $\mathcal{B}_i - \mathcal{B}_j$	DI: b_i/b_j	Ricci v. DeStefano
i	A	1	0	0	5	6	1/6	1/6	0	-0.06	0.38	Likely no bias
j	A	7	0	1	10	18	8/18	7/18	1/18			
i	B	0	1	0	5	6	0/6	1/6	-1/6	0.22	NaN (1 by context)	Bias for i
j	B	0	7	0	11	18	0/18	7/18	-7/18			
i	α	1	1	0	5	7	1/7	2/7	-1/7	0.23	2.71	Bias for i
j	α	1	7	0	11	19	1/19	8/19	-7/19			

the definition of the Objective Fairness Index in Definition 2.7. We provide a graphical representation in Figure 1.

DEFINITION 2.7. *The Objective Fairness Index is the difference between two groups’ marginal benefits.*

$$OFI = \mathcal{B}_i - \mathcal{B}_j = \frac{FP_i - FN_i}{n_i} - \frac{FP_j - FN_j}{n_j} \in [-2, 2] \quad (5)$$

2.3 Improvement over Disparate Impact

To illustrate the Objective Fairness Index’s significance, let us consider the instances in Table 1, similar to the case of New Haven and its firefighters. The same objective test is given in all scenarios. In Scenario A, the employer fulfills their promise and promotes (OFI=-0.06, DI=0.38). In Scenario B, the test is discarded (OFI=0.22, DI=1). Note Scenario B’s DI score is undefined. However, since the philosophy of DI suggests equal positive prediction rates indicate no bias, we use DI = 1.

With OFI’s absolute value being greater in Scenario B than in Scenario A, OFI is consistent with the ruling of Ricci v. DeStefano (2009). However, DI disagrees with this precedent, suggesting for strong bias in Scenario B and none in Scenario A.

In Scenario α , we conduct a thought experiment with a smoothing technique. Scenario α adds one qualified individual to each group that benefits (e.g., is correctly promoted) to Scenario B. Adding this single correct prediction changes DI’s score from 1 to 2.71, suggesting a transition from no bias to heavy bias for group i . However, OFI adjusts from 0.22 to 0.23 in the same Scenario α , illustrating its robustness.

Unlike DI, the Objective Fairness Index indicates the correct directional bias in both scenarios. Additionally, OFI’s defined-everywhere property and robustness against small changes showcase its stability. However, we note that DI is complementary to OFI in systemic disparity detection. With OFI, we can identify if an algorithm is at fault (OFI $\neq 0$) or if a systemic disparity exists outside of the algorithm (OFI ≈ 0 and DI $\neq 1$).

2.4 Establishing OFI Thresholds

Often, data has noise and perfectly objective tests are an open question. As such, we use thresholds. Some thresholds are given by a rule of thumb, such as DI’s Four-Fifths Rule. However, we formally derive OFI’s threshold to an upper bound of $OFI \in [-0.3, 0.3]$. We obtain these bounds by calculating the standard deviation of OFI over all confusion matrices (see the proofs in our code repository). We stress that the threshold ought to be reduced when we

expect better than uniform confusion-matrix distributions, such as in predicting recidivism.

3 Applying OFI: Empirical Case Studies

We explore the application of the Objective Fairness Index in practical scenarios: COMPAS (predicts recidivism), and Folktabels’ Adult Employment dataset (predicts employment).

For COMPAS, we obtain the confusion matrix from actual COMPAS predictions and cases of recidivism occurring within two years [1, 14]. Since we assume the positive label is beneficial, we flip the COMPAS positive label to be “not a recidivist”.

Figure 2 shows bias scores for each ethnicity pair in the COMPAS dataset. Here, we see that OFI affirms DI’s findings. This is an important revelation, as some argue that the disparate prediction rates are correct. In return, some argue that the labeled dataset suffers from systematic bias. However, this case illustrates that OFI can sidestep the need to prove incorrect ground labels, and shows that even if one assumes correct ground labels, there is algorithmic bias against certain races in COMPAS (OFI $\neq 0$). We believe this revelation will encourage more people to take corrective measures.

For Folktabels’ dataset, we analyze data from the Georgia census (393,236 observations) to predict employment status using a Random Forest classifier (RF) and Naïve Bayes (NB). In RF, OFI corroborates the findings from DI. However, OFI reveals differences to DI in the NB experiment (see Figure 3). Here, DI suggests a positive bias for Pacific Islanders in 7/8 cases. In contrast, OFI shows positive bias in 2/8 cases.

With the lens of objective testing, we find that Pacific Islanders suffer from algorithmic bias, despite having high disparate impact scores. These studies demonstrate OFI confirming suspicions of algorithmic bias (COMPAS), and when protected classes suffer from algorithmic bias, despite a high positive prediction rate.

4 Conclusions and Future Directions

In this work, we introduce the Objective Fairness Index (OFI) to evaluate bias in a legally grounded and context-aware perspective. By leveraging legal frameworks and precedents, particularly those stemming from objective testing and disparate impact theory, OFI captures the nuanced understanding of bias that encompasses both what occurred and what ought to have occurred.

In Section 2, we illustrate that the Objective Fairness Index is legally grounded, even where the disparate impact metric fails. Specifically, OFI finds no bias in objective testing while DI may or may not, and OFI can correctly detect bias where DI does not.

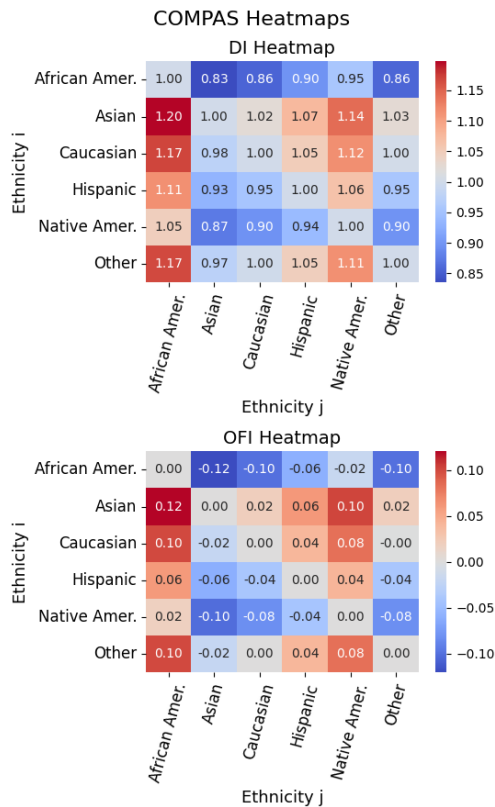


Figure 2: OFI shows that COMPAS manifests algorithmic bias in addition to manifesting disparate impact (DI). We use the predictions and labels published in [1].

Our analyses include legal-case-driven situations and popular bias problems, including COMPAS (risk of recidivism).

The Objective Fairness Index addresses a significant gap in the machine learning and AI fairness literature by offering a metric that not only aligns with legal standards but also provides novel insights into binary classification problems. Our findings highlight the limitations of existing metrics, such as disparate impact, which lack context and may lead to misinterpretation or oversight of critical bias factors. OFI offers a solution by integrating objective testing directly into the evaluation of bias.

For future work, we envision OFI in more complex scenarios, including multi-class classification problems, regression tasks, and recommender systems. Another promising avenue is the exploration of OFI's role in the development of debiasing techniques.

Additionally, corrective procedures should be investigated. As an objective measure, OFI identifies discriminatory selections, whereas reparative measures such as DI are for identifying social disparities. Thus, if the OFI fails, correction at the algorithmic or employer level is appropriate. However, if OFI shows no bias but reparative measures do, this suggests systematic disparity and can rationalize social programs. We recommend that all measures and corrective procedures are routinely scrutinized regarding ethics and law.

There are some inherent limitations to applying concepts in law to binary classification. Examples include the “good faith” of

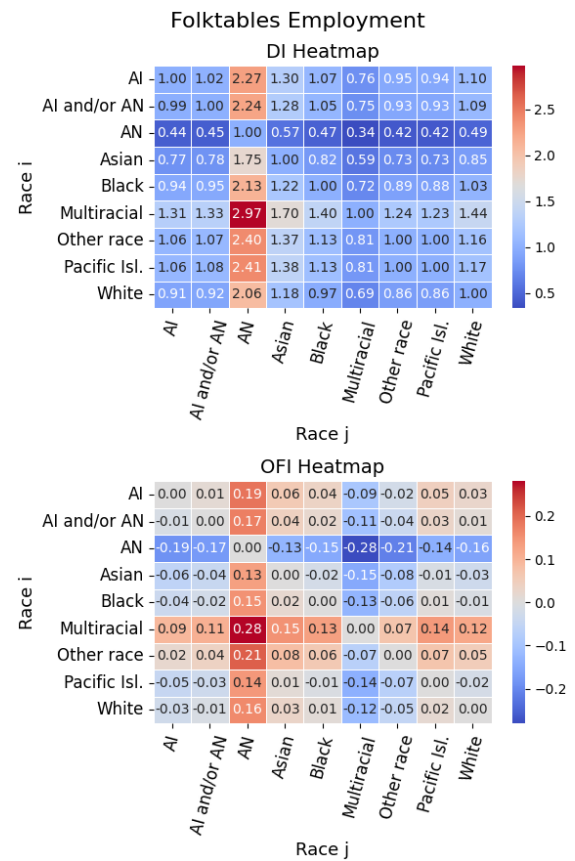


Figure 3: DI conveys that Pacific Islanders have positive bias for 7/8 comparisons. However, OFI shows that Pacific Islanders suffer from algorithmic bias in 6/8 cases. Key: AI is American Indian, AN is Alaska Native, Pacific Isl. is Pacific Islander.

the test-maker (Canada’s Meiorin test), proportional responses (UK’s Equality Act 2010, suggesting a non-binary decision), and expectations of “reasonable accommodations” for certain situations (Americans with Disabilities Act). We hope to explore solutions for these in more complex scenarios where they are addressable.

Our work underscores the importance of interdisciplinary approaches in tackling bias in AI. By grounding our metric in well-established legal precedents and demonstrating its applicability to contemporary machine learning challenges, we contribute a tool for researchers striving for fairness and transparency in AI systems.

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